Fiscal Year:	FY 2013	**Task Last Updated:**	FY 10/03/2013
PI Name:	Bassham, Diane Ph.D.		
Project Title:	Molecular Basis for the Role of the Vacuole in Shoot Gravitropism		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	50011-0001	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Biology--Microbial, Plant, Cell Biology: NNH08ZTT003N
Start Date:	08/01/2009	**End Date:**	07/31/2013
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 7/31/2013, per NSSC information (Ed. 5/8/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Task Description:	Plant perception and response to gravity is critical for appropriate growth, development, and environmental responses. Shoot gravitropism involves the perception of gravity by sedimentation of starch-containing amyloplasts in gravity-sensing cells, followed by transmission of the signal to additional cell types to promote directional growth. Vesicle trafficking to the vacuole is essential for the perception of the gravity signal, but the molecular basis for this requirement is unknown. Our long term goal is to determine the mechanism by which gravity is perceived by shoots and the gravity signal transduced to other cell types to produce a response. The objective of this application is to determine the function of vacuolar trafficking proteins in shoot gravitropism. We hypothesize that appropriate physical properties of the vacuole are essential for amyloplast sedimentation, and therefore for gravity perception. We will test this hypothesis by measuring the elasticity of wild type and mutant vacuoles using atomic force microscopy and by comparing the lipid constituents of the vacuolar membranes. We will also identify additional proteins involved in vacuolar trafficking by a combination of in vitro membrane fusion assays and in vivo immunoprecipitations and test their function in gravitropism using genetic approaches. This research will provide a molecular explanation for the observation that shoot gravitropism is specifically defective in vacuolar trafficking mutants and allow us to make testable predictions about the mechanistic basis for their agravitropic phenotype. It will also provide a foundation for the generation of transgenic plants with altered gravity response thresholds.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The primary impact of our research will be to increase understanding of the mechanisms by which plants respond to gravity. In addition to being important for the growth of plants in the altered gravitational conditions likely to be encountered during space flight, this research will also contribute to our knowledge of plant growth, development, and response to environmental cues. The gravitational response interacts extensively with other developmental, hormonal, and environmental response pathways, leading to morphological and anatomical changes to the plant structure. Understanding of the molecular mechanism of gravitropism is therefore vital for the development of strategies to improve plant growth and yield under varying conditions. The development of technologies to analyze the physical properties of vacuoles and in particular the fusion between membranes is also likely to impact research in plant cell structure, stress responses, and growth and development under different conditions.
Task Progress:	Personnel Rahul Roy (PhD student) continues to work full time on this project. A PhD student, Xiaochen Yang, is now also working on the project. Three undergraduate students have assisted in routine aspects of the research. Research progress Our focus in the past year has been continued characterization of two vacuolar trafficking mutants that we demonstrated are required for correct growth responses to gravity. Our goal is to understand the mechanistic basis for the gravitropic phenotype and its connection to the vacuolar trafficking defects. 1. TNO1. We showed previously that the TNO1 protein is required for efficient transport of proteins to the vacuole and initiated a characterization of the gravitropic defect of a tno1 mutant. Dark grown hypocotyls of tno1 knockout mutants bend slower than wildtype and TNO1 complemented lines when gravistimulated. In the past year we have made several significant discoveries related to this phenomenon: (a) Analysis of gravitropic bending kinetics in dark grown hypocotyls indicates that the knockouts show a significant lag in bending kinetics compared to WT and complemented lines. (b) TNO1 is also involved in root gravitropism. This was unexpected, as root and shoot gravitropism occur by largely distinct mechanisms and there is little overlap in the components required for each. (c) tno1 mutant roots also show a growth pattern that is skewed from the vertical, indicating a defect in plant growth or root morphology. (d) We hypothesized that these root and shoot gravitropic and growth defects could be due to either disruption of auxin fluxes or disorganization of microtubules. In support of this, the tno1 knockout mutants are hypersensitive to microtubule drugs, both in terms of root skewing, hypocotyl elongation, and cell morphology. Mutant roots were also significantly more sensitive to a synthetic auxin, 2,4D. (e) We are now in the process of visualizing auxin gradients and microtubule structure using fluorescent markers. (f) Electron microscopy of Golgi stacks in tno1 mutants has allowed quantification of defects in Golgi and trans-Golgi network structure between wild-type and mutant plants, providing insight into the cellular function of TNO1. 2. VPS45. VPS45 is a vacuolar trafficking protein that we have previously characterized. However, a knockout mutant in VPS45 is lethal, making the study of its role in gravitropism difficult. We recently identified a point mutation in the VPS45 gene via TILLING that causes a single amino acid change in the protein sequence. The mutant has a weak phenotype and has defects in both shoot and root gravitropism as well as general growth and morphological defects. We have complemented this mutant phenotype using a genomic fragment containing VPS45 under its own promoter, indicating that the mutant phenotypes seen are indeed due to the change in VPS45. Further analysis of the gravitropism defects is underway. 3. SYP51/52. A homozygous mutant in which expression of the t-SNARE SYP51 is reduced was identified. Initial gravitropism assays showed no significant shoot gravitropism defects, possibly due to functional redundancy or the residual SYP51 expression. Due to these difficulties, we are taking a different approach of using RNA interference to known down expression of SYP51 and SYP52 simultaneously. Constructs have been generated and introduced into Arabidopsis plants and transformants are in the process of being screened for reduced expression. Vacuolar properties. We hypothesize that changes in the physical properties of vacuoles contributes to shoot gravitropism defects in vacuolar trafficking mutants, either due to differences in protein content or in lipid composition. To test this hypothesis, we isolated vacuoles from wild-type Arabidopsis plants and from a vacuolar trafficking mutant (vti11) that has shoot gravitropic defects. Lipids were extracted from the vacuoles and sent to the Kansas State lipidomics facility for analysis. The data obtained were inconsistent, most likely because of the low concentrations of lipid in the samples. These experiments are in the process of being repeated with larger lipid quantities.

Bibliography Type:	Description: (Last Updated: 09/18/2019)
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